

# Research Skills

Grade 8 Science

October 28, 2025





# Agenda

- Finding and evaluating sources
  - Types of sources and where to find them
  - Evaluating sources
  - Exercise
- Citation crash course
  - Why citations matter
  - MLA style
  - Exercise
- AI in research
  - Do's and don'ts
  - CHAT with AI



# Researching in Science

- Science isn't just about knowing things, it's about **proving** it
- Research is how you find the **real answers** and build a case for your claims
- Think of yourself as a detective **finding, evaluating, and applying** evidence to support an argument



Fig. 1. "Research Process." *The Research Process*, Forsyth Tech Community College, 15 July 2025, <https://forsythtech.libguides.com/researchprocess>.

# Finding and evaluating sources

Encyclopedias, Academic Books, Reputable Websites



# Research with Encyclopedias

Get the big picture! Quickly learn the most important facts, definitions, and dates

## TIPS:

- Find key terms: Learn the right vocabulary or “**keywords**” to use when you search in books or websites (e.g. energy; kinetic)
- Follow the clues: Most entries list their sources or have an “Additional Reading” section at the end



# Research with Academic Books

Find the most detailed information available on your topic,  
written by experts in the field

## TIPS:

- **Don't read the whole book:** use the **Table of Contents** at the start to find the right chapter, and use the **Index** at the back to look up specific keywords
- **Skim** - once you find the right page, scan the **headings** and **first sentences** of paragraphs to see if they are relevant

# Research with Websites

Certain websites offer valuable information for research but generally, websites must be explored with caution!

## TIPS:

- Pay attention to domain type:
  - Anything: **.com, .ca, .org**
  - Government: **.gc.ca OR .gov**
  - Educational institutions: **.edu**
  - Intergovernmental: **.int**
- Search Google using:
  - **Keywords** (muscle; growth; workout)
  - Place phrases into **quotation marks** (“cardiovascular system”)

**DO NOT TRUST THE “AI OVERVIEW” PROVIDED BY GOOGLE - IT IS OFTEN WRONG**



# Look here for reliable sources:

## Encyclopedias

- Reliable overview with key facts, dates, and terms
- Example:
  - Encyclopedia Britannica
    - Username: halifaxgs
    - Password: seven7eight8

## Academic Books

- Written and fact-checked by experts, offer most detailed information
- Examples:
  - Google Books
  - Visit the school library (upstairs), ASK ME FOR HELP FINDING MATERIALS

## Reputable Websites

- Official sites from governments, universities, and scientific organizations, run by experts with trusted information
- Examples:
  - Canada Health Infobase
  - National Geographic Education
  - Medline Plus

# Evaluating Sources - CRAAP Test

## Currency

- Is the information up-to-date for your topic?
  - Look for published or “last updated” dates

## Relevance

- Does the information relate to your topic?
- Is it at a good reading level for you/can you understand it?

## Authority

- Who is the author (or organization) and are they qualified to write about the topic?
  - Look for author name and look them up!
  - Pay attention to domain

## Accuracy

- Does the language seem unbiased and free of emotion?
- Are there spelling or grammatical errors?

## Purpose

- What is the reason for the site? Is it to inform and teach or is it designed to sell, entertain, or persuade?



# Website Evaluation Exercise

In pairs, apply the CRAAP method to your assigned website (click link to the right) and complete Part 1 of your worksheet.

**Research question:** Is there scientific proof that aliens exist?

Website A - [Aliens Info](#)

Website B - [NASA](#)

**STAY ON THE  
WEBPAGES THAT OPEN**

# Citing in your research



## Why cite?

- Says "thank you" to the experts whose work you used
- Using someone else's ideas or words without giving them credit is plagiarism (stealing)
- Allows your teacher and others to find your sources, proving you didn't make things up
- Creates a trail for you to trace your work back

## When should you cite?

- You must cite a source when you use:
  - Research
  - Ideas
  - Quotes
  - Data
  - Images, graphics, videos, or other media

***that are not your own, INCLUDING AI***

# MLA Style

MLA (Modern Language Association) style gives you a standard way to credit your sources

It has two main parts: **In-text citations** and a **Works Cited list**

**\*Every source type has different COMPONENTS/FORMAT/PUNCTUATION**  
(books, websites, videos, etc.)

Its core elements may include:

- Author name
- Title of book/web page
- Publisher
- Publication/Last modified date (or (n.d.) for no date)
- Location/URL

# MLA - In-Text Citation

A quick note inside your paragraph showing where the info came from

| Type                                          | Description                                            | Example                                                                                                    |
|-----------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Books                                         | Includes the author's last name and page number        | The Earth's crust is made up of tectonic plates (Smith 12).                                                |
| Website with Author                           | Includes just the author's last name                   | The process that allows plants to create their own food is called photosynthesis (Perkins).                |
| Website with Organization as Author/No Author | Just list the title of the web page in quotation marks | The Mars Rover has traveled over 20 kilometers on the planet's surface ("Explorations of the Mars Rover"). |



# MLA - Works Cited

A full alphabetical list at the end of your project with all the details for every source you used

| Type                                        | Format                                                                                                                                                          | Example                                                                                                                                                                                                         |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Book                                        | Author's Last Name, First Name. <i>Title of Book</i> . Publisher, Publication Year.                                                                             | Smith, John. <i>The Science of Volcanoes</i> . Geology Press, 2019.                                                                                                                                             |
| Web page - with author                      | Author's Last Name, First Name. "Title of the Page or Article." <i>Name of the Website</i> , Publication or Last Modified Date, URL. Access date (optional).    | Perkins, Jane. "Photosynthesis." <i>Science Education</i> , 16 June 2024, <a href="http://www.scienceeducation.com/photosynthesis">www.scienceeducation.com/photosynthesis</a> . Accessed 22 October 2025.      |
| Web page - organization as author/no author | "Title of the Page or Article." <i>Name of the Website</i> , Sponsoring Organization/Publisher, Publication or Last Modified Date, URL. Access date (optional). | "What Is the Greenhouse Effect?" <i>NASA Climate Kids</i> , NASA, 6 Oct. 2022, <a href="http://climatekids.nasa.gov/greenhouse-effect/">climatekids.nasa.gov/greenhouse-effect/</a> . Accessed 22 October 2025. |

# Citing AI

- You must cite your use of AI anytime it is paraphrased, quoted, or incorporated into your work in any way (including brainstorming or editing)

## Works Cited List

"Description of your prompt" prompt. *Name of AI Tool*, Version, Name of Company, Date you got the answer, General URL.

"Explain the differences between a plant cell and an animal cell for an 8th grader" prompt. *ChatGPT*, GPT-5, OpenAI, 21 Oct. 2025, chat.openai.com.

## In Text Citation

Plant cells get their strong, box-like shape from a cell wall, but animal cells only have a flexible cell membrane ("Explain the differences").

# Citation Generators

- If your teacher permits you to use citation generators, they are generally fine to use
- Keep in mind they make mistakes MOST of the time
- Use a reliable generator such as **ZoteroBib**
- Build a **foundation** with the generator and **fix/adjust** as necessary

## Tip:

- Be cautious of ones that have lots of advertising, provide little information on who is behind it, or want you to download software (potentially malware)



# Citation Exercise

On your own, try creating a citation for this website.  
Fill in Part 2 of the worksheet.

**CLICK HERE TO**  
**OPEN THE WEBSITE**

“Necrotizing Fasciitis (Flesh-Eating Disease).” *HealthLink BC*, British Columbia, 19 February 2019. [www.healthlinkbc.ca/healthlinkbc-files/necrotizing-fasciitis-flesh-eating-disease](http://www.healthlinkbc.ca/healthlinkbc-files/necrotizing-fasciitis-flesh-eating-disease). Accessed 28 Oct. 2025.

# GenAI in Research



# GenAI in Research

## DO'S

- **DO** use it to brainstorm topics and research questions
  - “Give me 5 project ideas about Newton's Laws of Motion”
- **DO** ask it to explain complex ideas in simple terms
  - "Explain cellular respiration like I'm an 8th grader"

## DON'TS

- **DON'T** copy and paste - this is plagiarism (cheating) and it doesn't show what you learned
- **DON'T** trust everything it says - AI can be wrong or "hallucinate" (make up facts)
- **DON'T** use it as your only source, you must find real, citable sources

# How to **CHAT** with AI: *Better Prompts = Better Answers*

## **C – Choose your goal**

Decide what you really want to find out or create

## **H – Help it help you**

Give details, examples, or context so the AI knows what you mean (“I am a Grade 8 science student working on a research project”)

## **A – Ask clearly**

Use full questions or instructions — be specific! (“List 5 environmental impacts of climate change in Nova Scotia”)

## **T – Try again**

Look at the answer. If it’s not quite right, tweak your question and try again



# ADDITIONAL RESOURCES

## Citations

1. [Owl Purdue MLA Formatting and Style Guide](#)
2. [Citing Generative AI in MLA Style](#)
3. [ZoteroBib Citation Generator](#)



# Thank you

Stop by my desk in the library anytime!  
Or email me at [sbateman@halifaxgrammar.ca](mailto:sbateman@halifaxgrammar.ca) with questions